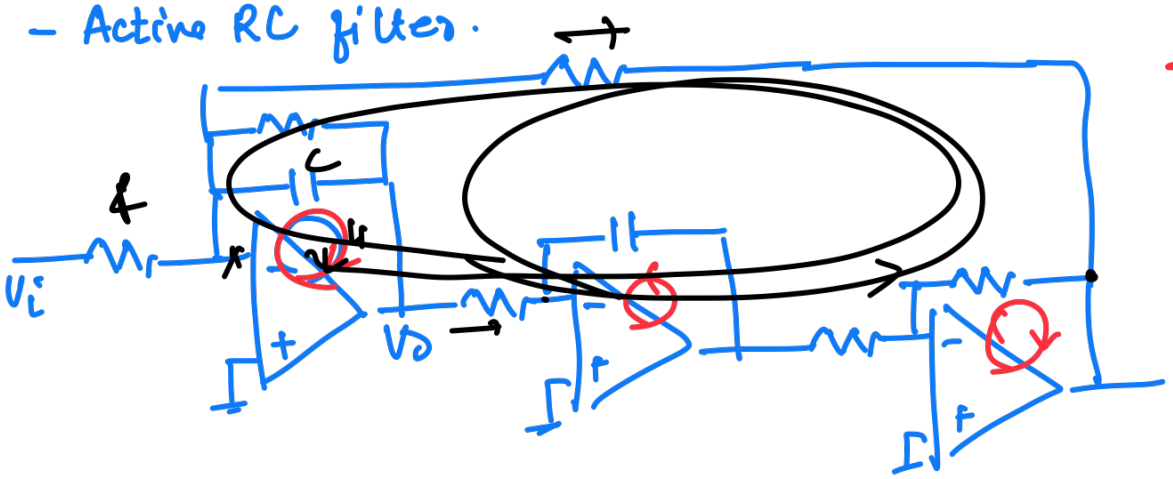
Lecture #30

- Active RC filter.



$\frac{V_o}{V_i} =$ ( )

- PSTB (Periodic Stability Analysis)
- STB (Stability analysis)





$R = \rho \frac{L}{A}$



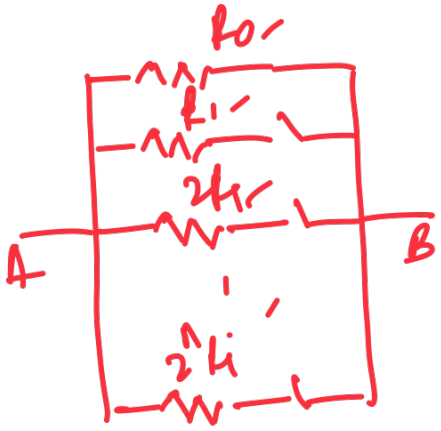
$\frac{1}{T}$

- $R_0 \pm 25\%$
- $C_0 \pm 35\%$



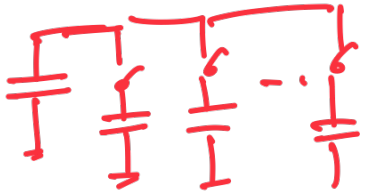


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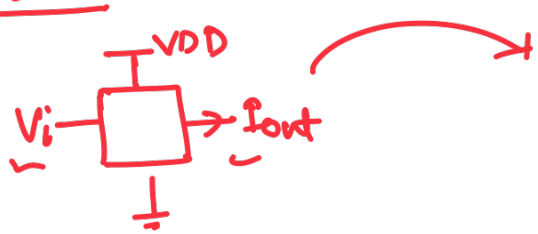
Resistor Bank.





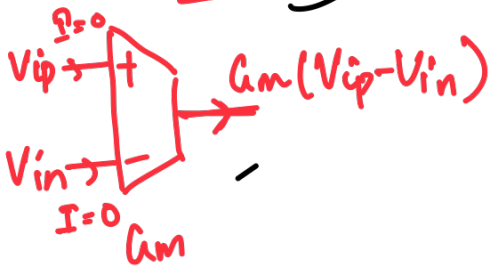
Capacitor Bank

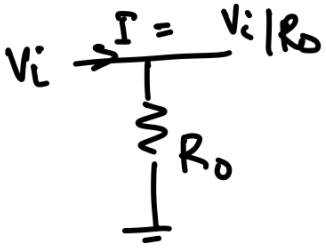
Gm Block.

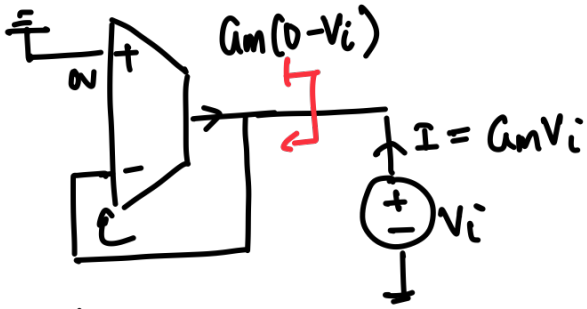




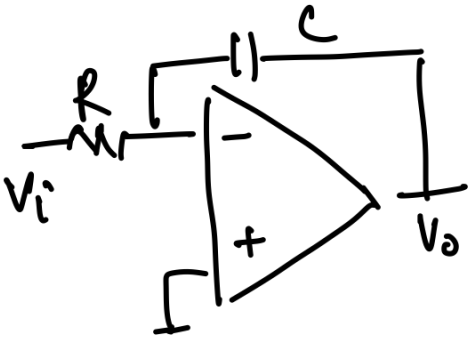




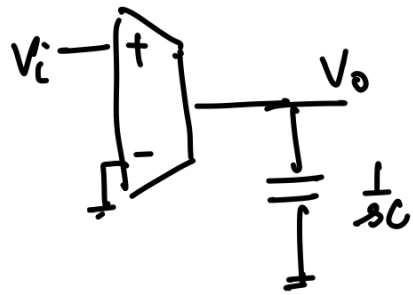




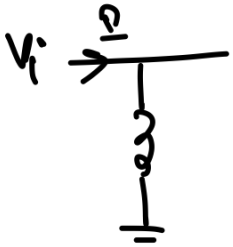
$$Z = \frac{V_i}{G_m V_i} = \frac{1}{G_m}.$$



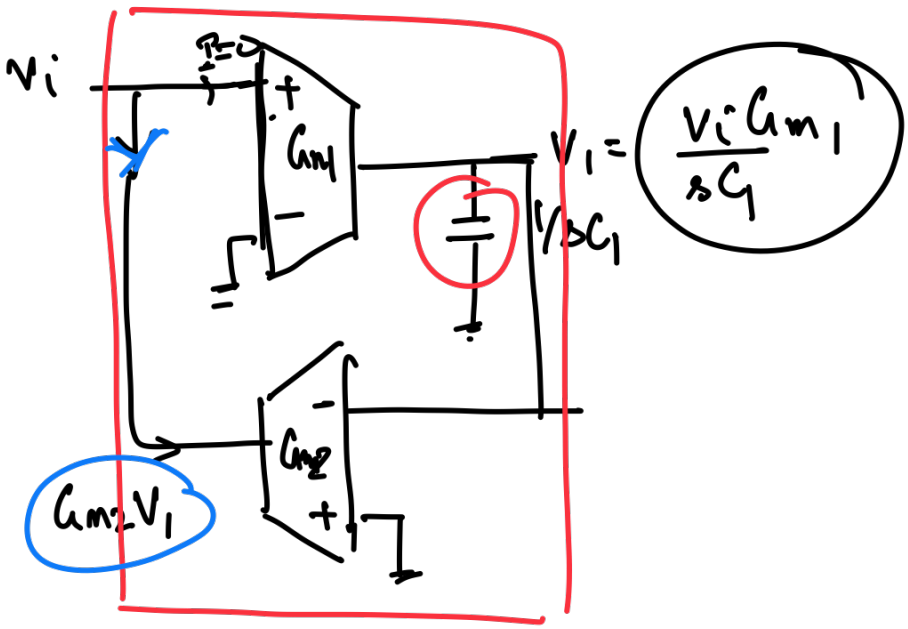
$$\frac{V_o}{V_i} = -\frac{1}{sRC}$$



$$V_o = \frac{G_m V_i}{sC} = \frac{V_i}{sC/G_m}$$



$$I = \frac{V_i}{sL}$$

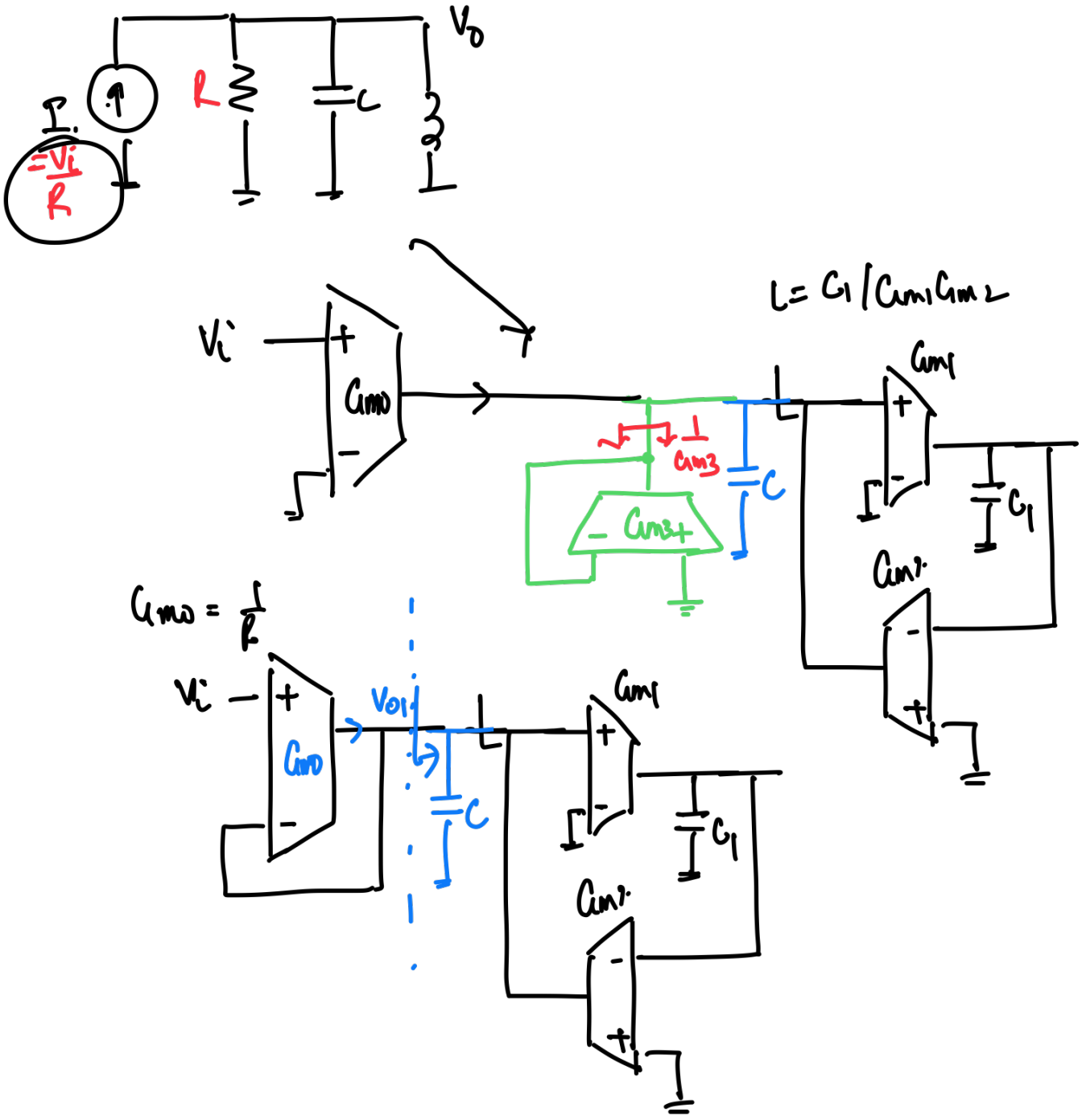


$$I = G_{m2} V_1 = G_{m2} \frac{G_{m1} V_i}{sC_1}$$

$$= \frac{V_i}{s\left(C_1 / G_{m1} G_{m2}\right)}$$

$$Z = \frac{V_i}{I} = s\left(\frac{C_1}{G_{m1} G_{m2}}\right)$$

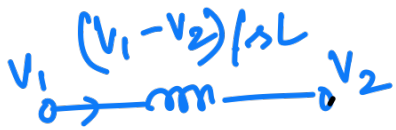
$$\simeq L$$

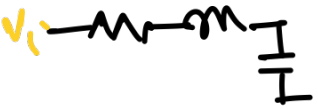


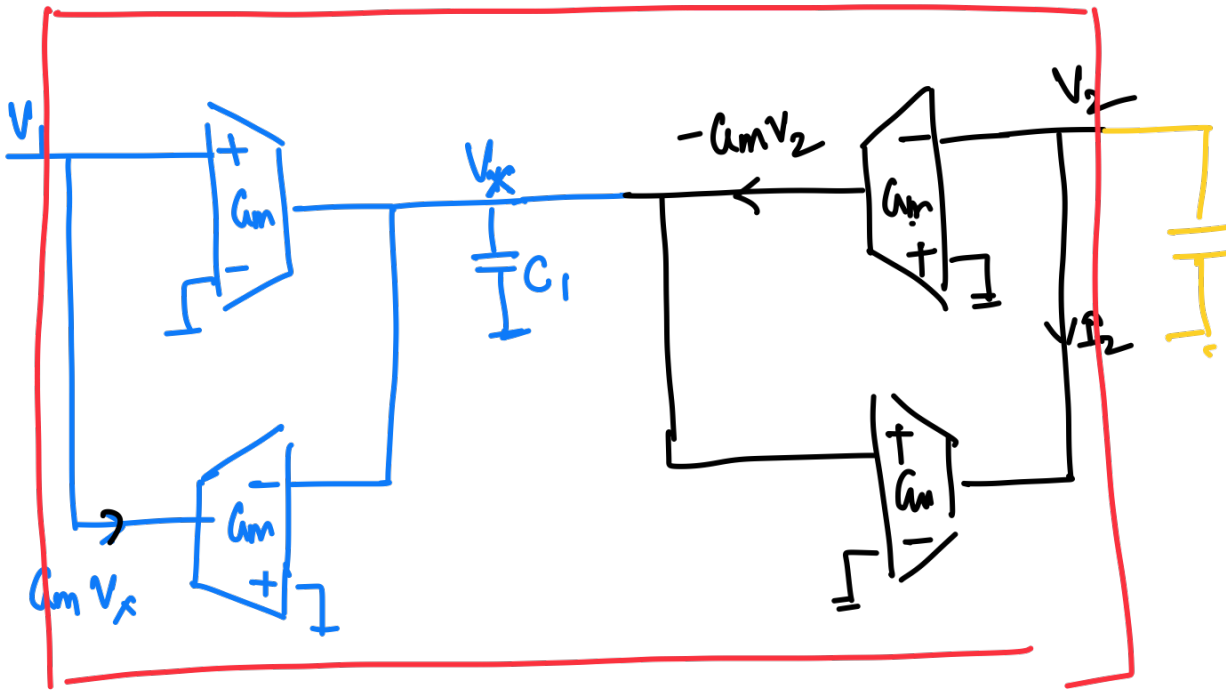
$$I = G_{mo}(V_i - V_{o1}) = \frac{V_{o1}}{(sL \,||\, \frac{1}{sC})}$$

$$\Rightarrow G_{mo}V_i = G_{mo}V_{o1} + \frac{V_{o1}}{sL \,||\, \frac{1}{sC}}$$

$$= \frac{V_{o1}}{1/G_{mo}} + \frac{V_{o1}}{sL} + V_{o1}.sC$$







$$V_x = \frac{G_m V_1}{sC_1}$$

$V_x \propto (V_1 - V_2)$

$$V_x = (G_m V_1 - G_m V_2)\frac{1}{sC_1}$$

$$= (V_1 - V_2)\frac{1}{sC_1/G_m}$$

$$I_1 = G_m V_x = \frac{G_m(V_1 - V_2)}{sC_1/G_m} = \frac{(V_1 - V_2)}{sC_1/G_m^2}$$